NGF COLLEGE OF ENGINEERING & TECHNOLOGY

MID-TERM EXAMINATION 2026

B.TECH (CSE/ AI/ML)

DATA STRUCTURES & ALGORITHMS

Max Marks: 30 **Time:** 1.5 Hours

Section A – Compulsory Short Answer Questions 3 × 2 = 6 Marks

1. In a doubly linked list, what is the purpose of the prev pointer?
2. What is the time complexity of accessing the i-th element in a singly linked list?
3. How many fields does a node in a singly linked list typically contain?
4. What is the basic unit of a singly linked list?

Section B – Attempt any three of the rest of the questions 24 Marks

1. Explain the purpose of visited nodes in a depth-first traversal of a graph.
2. Describe what a graph cycle is and why directed graphs may or may not contain them.
3. What is the time complexity of Dijkstra's algorithm using a binary heap?

***** End of Question Paper *****